

COMPUTER NETWORKS

Week-8

5. Network Layer

- The Network Layer performs
 - network routing functions,
 - report delivery errors.
- Routers operate at this layer—sending data throughout the extended network and making the Internet possible
- Decide which path (route) the data packet should take
- Ensure it reaches the correct destination address (IP address)

Network Layer

- Checks destination IP
- Chooses the best route via routers
- Forwards packets step-by-step until they reach the server

Network Layer

- Devices called routers operate at the Network Layer
 - Read the destination IP
 - Decide the next hop
 - Forward the packet
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- The Network Layer finds the best path and delivers packets from one **network** to another using IP addresses

6. Data Link Layer

- Deliver data within the same network
- Use MAC addresses to identify devices
- Ensure error-free delivery between directly connected devices
- The Data Link Layer ensures reliable delivery of data between devices on the same network using MAC addresses
- Data Link Layer moves data between neighboring devices, while Transport Layer ensures complete and reliable delivery between the sender and receiver.

Data Link Layer

- Data Link Layer works at the hardware level, so it needs a hardware identifier (MAC)
- IP address gets the data to the correct network, but MAC address delivers it to the exact device inside that network

7. Physical Layer

- Provides physical interface for transmission of information.
- Defines rules by which bits are passed from one system to another on a physical communication medium.
- Covers all - mechanical, electrical, functional and procedural - aspects for physical communication.
- Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are defined by physical layer specifications.
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Physical Layer

- The Physical Layer is responsible for actually sending raw bits (0s and 1s) as signals over a medium

Physical Layer

- The Physical Layer deals with:
- Cables (Ethernet, fiber)
- Voltage levels (how 0 and 1 are represented)
- Data transmission speed
- Connectors (RJ45, etc.)
- It does not understand data, only signals

Physical Layer

- Suppose you send a file:
- Other layers prepare the data
- Data Link Layer creates a frame
- Physical Layer:
 - Converts frame into bits (0s and 1s)
 - Converts bits into electrical signals (in a cable)
 - Transmits them over the wire
- On the receiving side:
 - Signals → bits → data (reverse process)
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Data Frame

- Created at Data Link Layer
- Contains:
 - Source MAC
 - Destination MAC
- Used for delivery within a local network
- Switches read frames

Important ..

- When data is sent:
- Transport Layer → creates segments
- Network Layer → adds IP → creates a packet
- Data Link Layer → adds MAC → creates a frame
- Physical Layer → sends bits
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- Routers read packets
- Switches read frames